STAT7630: Applied Linear Models

Final Project Guidelines

Due Date: May 5, 2026, by 11:59 PM

Overview

The final project is an opportunity to apply the methods from STAT7630 to a real-world dataset of your choosing. You will independently identify a dataset, formulate meaningful research questions, conduct a full regression analysis, and communicate your findings in a formal written report.

The project is intended to assess your ability to carry out a complete applied statistical analysis from beginning to end, including:

- selecting an appropriate dataset and research question
- performing exploratory data analysis
- choosing and justifying regression models
- fitting and comparing models
- conducting diagnostic checks
- performing statistical inference
- interpreting results in context
- communicating findings clearly and professionally

This is not simply a coding assignment. The emphasis is on **statistical reasoning, model justification, interpretation, and communication**.

Project Requirements

Dataset

You must identify and use your own **real-world dataset**.

Acceptable sources include:

- public government datasets
- Kaggle
- UCI Machine Learning Repository
- published supplementary datasets
- R packages containing real data
- domain-specific open data repositories

The dataset should contain enough observations and variables to support meaningful regression analysis.

Expectations for the Dataset

Your dataset should:

- involve a substantive real-world question
- contain at least one response variable suitable for regression modeling
- contain multiple potential predictors
- be sufficiently rich to allow thoughtful modeling decisions
- not be a toy dataset created solely for demonstration purposes

If you are unsure whether a dataset is appropriate, ask early.

Required Analysis Components

Your project must include the following elements.

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1. Exploratory Data Analysis (EDA)

Conduct an initial exploration of the dataset.

Possible components:

- summary statistics
- missing data assessment
- histograms / density plots
- scatterplots
- boxplots
- correlation analysis
- group comparisons
- transformations if needed

Your EDA should be purposeful and directly connected to later modeling decisions.

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2. Model Design

Clearly define:

- the response variable
- candidate predictors
- scientific motivation for including predictors
- possible confounders
- potential interactions
- nonlinear effects (if relevant)
- whether transformations are needed
- whether alternative model families are appropriate (linear, logistic, Poisson, etc.)

You should explain *why* the proposed model is reasonable.

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3. Model Fitting

Fit one or more appropriate regression models.

Examples may include:

- multiple linear regression
- logistic regression
- Poisson regression
- mixed effects models
- models with interactions
- models with polynomial terms
- spline-based models
- regularized regression (if appropriate)

You are not required to use the most advanced method possible. A well-executed standard model is preferable to an unnecessary complicated model.

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4. Diagnostics

Assess model adequacy using appropriate diagnostics.

Examples:

Linear Regression

- residual vs fitted plots
- Q-Q plot
- scale-location plot
- leverage / influence diagnostics
- multicollinearity checks

Logistic Regression

- calibration plots
- residual diagnostics
- leverage / influence
- discrimination metrics

Poisson Regression

- residual plots
- overdispersion checks
- fitted vs observed counts

Diagnostics should be interpreted, not merely displayed.

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5. Inference and Interpretation

Provide substantive interpretation of results.

Examples:

- estimated effects
- confidence intervals
- hypothesis tests
- effect sizes
- predicted outcomes / probabilities
- comparisons between groups
- practical significance

Interpret results in the context of the application, not only in statistical notation.

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6. Discussion

Reflect on the broader meaning of your findings.

Possible topics:

- substantive conclusions
- limitations of the data
- model assumptions
- possible bias or confounding
- alternative modeling choices
- future work
- recommendations

Written Report Format

Your report must be written in the following structure.

Title Page

Include:

- project title
- your name
- course name (STAT7630: Applied Linear Models)
- submission date

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1. Introduction

Describe:

- the application area
- the dataset
- the primary research question(s)
- why the question matters

This section should motivate the analysis.

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2. Methods

Describe:

- data source
- variables used
- preprocessing steps
- statistical methods
- model specification(s)
- diagnostic methods
- software used (R)

Write this clearly enough that another analyst could reproduce the workflow.

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3. Results

Present:

- EDA findings
- fitted model results
- tables of coefficients
- plots
- diagnostics
- statistical inference
- substantive interpretations

Use figures and tables effectively.

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4. Discussion

Summarize:

- key findings
- limitations
- implications
- possible extensions

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5. Supplementary Material

Include:

- additional plots
- extra model comparisons
- sensitivity analyses
- full code appendix (recommended)
- any supporting material not central to the main narrative

Length Guidelines

Suggested report length:

- **8–15 pages** main report (not including appendix/code)

Quality matters more than length.

Submission Requirements

Submit the following:

Required Files

1. Final report (PDF preferred)
2. R code used for analysis
3. Dataset or clear instructions for obtaining it

Submit through the course submission system.

Grading Criteria

Your project will be evaluated on the following dimensions.

Component	Weight
Quality of research question and dataset	10%
Exploratory data analysis	15%
Model design and justification	20%
Correct model fitting and diagnostics	20%
Inference and interpretation	20%
Quality of writing and presentation	15%

What Strong Projects Typically Do

Strong projects usually:

- ask a clear and interesting question
- choose an appropriate model
- justify modeling decisions
- include thoughtful diagnostics
- interpret results clearly
- communicate limitations honestly
- present clean, readable figures and tables

Common Mistakes to Avoid

- using a dataset that is too small or too trivial
- including many plots with no interpretation
- reporting output without explanation
- ignoring diagnostics
- overfitting without justification
- confusing statistical significance with practical importance
- poor organization or unclear writing

Suggested Timeline

By Mid-April

- Choose dataset
- Define research question

Late April

- Complete EDA and modeling

Early May

- Finalize report and polish writing

May 5, 2026

- Submit final project

Final Note

This project is your opportunity to demonstrate that you can think like an applied statistician: connect real data, statistical methodology, and scientific interpretation in a coherent analysis.

Choose a topic you find interesting and aim for clarity, rigor, and thoughtful conclusions.